UNIT III: Steering & Suspension System

Syllabus: Principle of steering, steering geometry & wheel alignment, steering linkages, steering gearboxes, power steering, front axle, suspension - independent & solid axle, coil / leaf / air suspension, torsion bar, shock absorbers.

1. Principle of Steering

- Steering system lets the driver guide the vehicle by turning the front (sometimes all four) wheels. Driver effort at the steering wheel is multiplied by the gearbox and transmitted through linkages to the wheels.

Functions

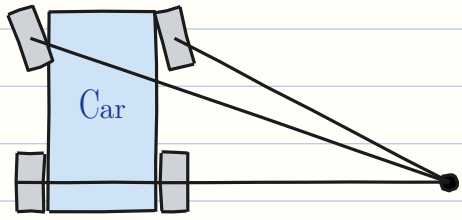
- Control direction of travel; provide directional stability; minimise tyre wear; give driver feel (road feedback); absorb shocks from road.

Requirements

- Easy & light operation, self-centring (return-ability), correct steering geometry, true rolling (no skid) of wheels on a turn.

Condition for true rolling - Ackermann principle

- During a turn all wheels must roll about one common point (instantaneous centre) on the line of the rear axle. Hence the inner wheel must turn more than the outer wheel.
- Condition: cot(phi) - cot(theta) = c / b (c = distance between pivot axes (track), b = wheelbase; theta = outer wheel angle, phi = inner wheel angle).



No skid -> no tyre wear
All wheel axes meet at I
inner wheel angle > outer angle

Fig 3.1 Correct steering (Ackermann condition)

Turning radius

Min. turning circle radius of a car ~ 4.5 - 6 m. Outer-wheel turning radius R = b / sin(theta) (approx.), where b = wheelbase.

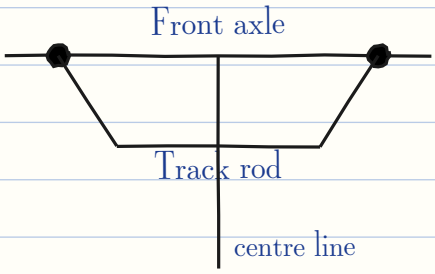
2. Ackermann & Davis Steering Gear

Davis steering gear

- Uses sliding pairs. Gives exact (mathematically correct) steering for all positions but has many parts that wear -> backlash; rarely used.

Ackermann steering gear

- Uses turning pairs only (simple, less wear). The track arms are inclined at angle alpha to the front axle and joined by the track rod. Only approximately correct - exact at straight-ahead and one more position, small error at others - but widely used.



Track arms inclined inwards,
lines through them meet at
a point on the rear axle,
centre of wheel base (approx.)
Used in all modern vehicles

Fig 3.2 Ackermann mechanism (top view)

Point	Davis	Ackermann
Pairs	Sliding (more wear)	Turning (less wear)
Accuracy	Exactly correct	Approximately correct
Maintenance	High	Low
Use	Rare	Universal

Types of steering by wheels

- Front-wheel steering (most vehicles), rear-wheel steering, four-wheel steering (4WS) - rear wheels turn opposite at low speed (tight turns) and same direction at high speed (stability); used in some SUVs / trucks.

Main parts of steering system

- Steering wheel -> steering column (collapsible / tilt) -> steering gearbox -> pitman arm -> drag link / tie rods -> steering arms -> stub axles -> wheels.

3. Steering Geometry - Camber & Caster

Steering geometry (wheel alignment angles) describes the angles of the front wheels & steering axis so that the vehicle is stable, steers easily and the tyres wear evenly. Main angles: camber, caster, king-pin inclination (KPI), toe-in & toe-out.

Camber

- Tilt of the wheel from the vertical when seen from the front. Positive camber - top outward; negative - top inward; zero = vertical.

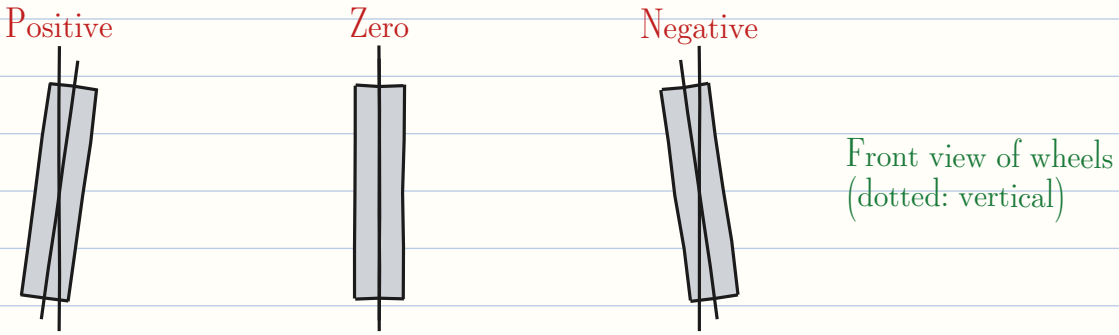


Fig 3.3 Camber angle

- Typical 0.5 deg - 1.5 deg positive. Purpose: (1) bring the tyre contact point under the king-pin -> reduces steering effort; (2) compensate for wear in bearings & flexing of axle under load; (3) prevent wheel falling out/tilt inward.
- Too much +ve camber -> outer tread wear; too much -ve camber -> inner tread wear; unequal camber -> vehicle pulls to side with more +ve camber.

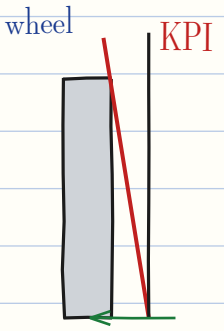
Caster

- Angle between the king-pin axis (steering axis) and the vertical, seen from the side. Positive caster - top of axis tilted backward (towards driver).
- Typical +2 deg to +4 deg (power-steering cars up to +6 deg). Purpose: directional stability + self-centring (return-ability) after a turn - like a caster wheel of a trolley / bicycle fork.
- Too much +ve caster -> heavy steering, shimmy; negative caster -> light steering but wander. Unequal caster -> pulls to side with less +ve caster.

4. KPI, Toe-in, Toe-out & Scrub Radius

King-pin inclination (KPI) / steering axis inclination (SAI)

- Inward tilt of king-pin top, seen from the front; 6 - 8 deg.



KPI: (1) reduces scrub radius -> less steering effort (2) returns wheel to straight ahead (lifts vehicle when wheel turned -> self-centring)

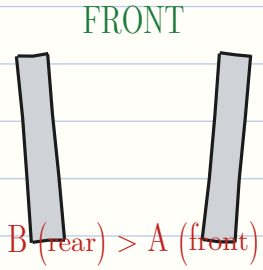
Included angle = Camber + KPI

Fig 3.4 King-pin inclination and scrub radius

- Scrub radius = distance between king-pin axis & tyre centre line at ground. Large +ve scrub -> kick-back on bumps; zero scrub -> less effort; negative scrub (modern FWD) -> stable braking.

Toe-in

- Front wheels are set so that the front edges are closer than the rear edges (distance A < B). Typical toe-in 2 - 4 mm (0 - 3 mm).



Toe-in = B - A

Purpose: offsets the tendency of +ve camber wheels to roll outward; keeps wheels parallel when rolling

Fig 3.5 Toe-in (top view)

- Toe-out on turns (turning angle): the inner wheel turns more than the outer (Ackermann) giving ~ 20 deg inner vs 18 deg outer - also tests bent arms.
- Wrong toe -> feathered tyre edges, rapid tread wear, wander.

5. Wheel Alignment & Balancing

- Wheel alignment = setting camber, caster, toe (and thrust angle) to manufacturer specification so that wheels are parallel / properly angled to each other and to the vehicle frame.

Symptoms of poor alignment

- Vehicle pulls to one side; steering wheel off-centre; uneven / rapid tyre wear; heavy or loose steering; vibration / shimmy; squealing tyres in turns; poor return-ability.

Causes

- Hitting kerb / pothole, worn ball joints, tie-rod ends or bushes, bent arms, sagging springs, changed ride height, accident damage.

Alignment procedure (4-wheel)

1. Check tyre pressure, wear and wheel run-out; inspect suspension & steering joints for play.
2. Place vehicle on level alignment rack; fit sensor heads / clamps (laser / camera / CCD) to all wheels; compensate run-out.
3. Read caster, camber, KPI, toe on the screen versus manufacturer data; adjust using eccentric bolts, shims, tie-rod sleeves.
4. Set toe last, centre the steering wheel, road test.

Wheel balancing

- Uneven mass around tyre-wheel assembly -> vibration. Static balance (single plane: wheel hops) & dynamic balance (two planes: wheel wobbles / shimmy). Corrected using lead / clip-on weights on rim edges, found on a computerised balancer.

Angle	Typical value	Main effect
Camber	+0.5 to +1.5 deg	Tyre wear, pull, steering effort
Caster	+2 to +4 deg	Stability, self-centring
KPI / SAI	6 - 8 deg	Self-centring, scrub radius
Toe-in	2 - 4 mm	Tyre wear, straight running

6. Steering Linkages

- The linkage transmits motion from the steering gearbox to the two steering knuckles so that both wheels turn through the correct angles.

Components

- Pitman arm (drop arm) - swings from the gearbox cross shaft.
- Drag link (steering link) - pitman arm to steering arm (long link).
- Steering arm / knuckle arm - fixed to stub axle (knuckle), rotates it on king-pin.
- Tie rod (track rod) with adjustable sleeve for toe setting; tie-rod ends (ball joint).
- Idler arm - supports the centre link (parallelogram linkage).
- Steering column & universal joint, collapsible (energy-absorbing) for safety.

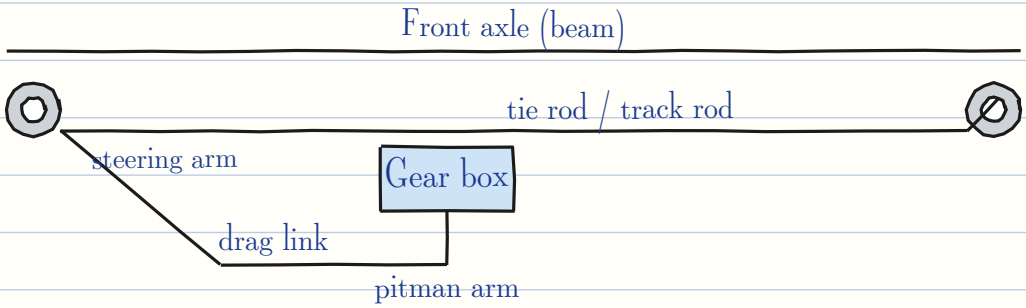


Fig 3.6 Steering linkage for a rigid-axle vehicle

Linkage layout types

- Rigid axle (solid axle): one-piece tie rod + drag link (above).
- Independent suspension: split (three-piece) tie rod - centre link + 2 tie rods + idler arm; or rack & pinion where tie rods are directly joined to the rack ends.
- Ball joints & rubber bushes need lubrication / replacement; worn joints cause free play and wander.

Steering ratio

Steering ratio = steering-wheel rotation / road-wheel swivel angle: 12:1 to 24:1. Higher ratio = lighter effort but more turns lock-to-lock (3-4 turns for manual, 2-3 for power steering).

7. Steering Gearboxes - Types

- The steering gearbox (a) converts rotary motion of the steering wheel into linear / arc motion of linkage; (b) gives mechanical advantage (gear ratio 12:1 - 24:1); (c) must be irreversible or semi-reversible (road shocks not felt at wheel).

Common types

Type	Mechanism	Use
Worm & sector	Worm gear meshes with a sector gear	Older cars / trucks
Worm & roller	Roller on cross-shaft rolls on worm (less friction)	Light trucks
Cam & lever	Cam (worm) with studs / lever bolt	Cars (Ross type)
Recirculating ball (nut)	Balls roll between worm & nut	Trucks, SUVs, buses
Rack & pinion	Pinion drives toothed rack	Most modern cars (FWD)

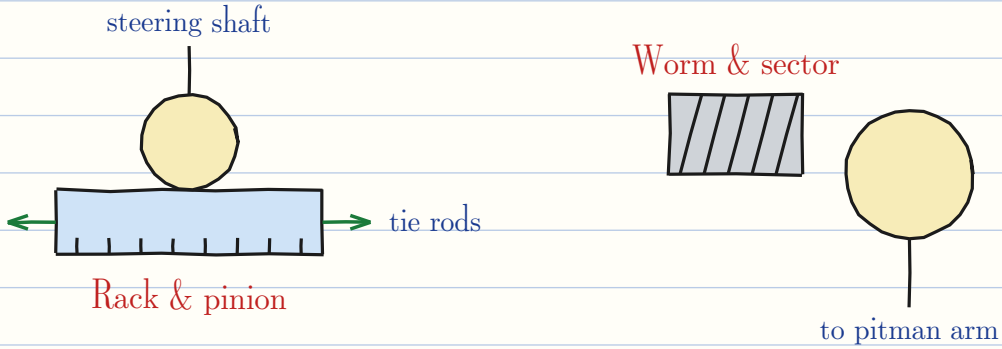


Fig 3.7 Rack & pinion and worm & sector

- Steering ratio = (steering wheel angle) / (pitman arm angle). Lower ratio -> quick response (sports cars); higher ratio -> less effort (heavy vehicles).

8. Recirculating Ball & Rack-Pinion

Recirculating ball steering gear

- A worm shaft with spiral groove carries a ball nut with matching groove; steel balls run in the groove, reducing friction (efficiency ~ 85-90 %). Balls leaving one end are returned through ball return guides (tubes) - recirculate.
- Ball nut has rack teeth meshing with a sector shaft -> sector (cross) shaft rotates -> pitman arm. Adjusters: end-thrust (worm bearings) and sector-lash screw.

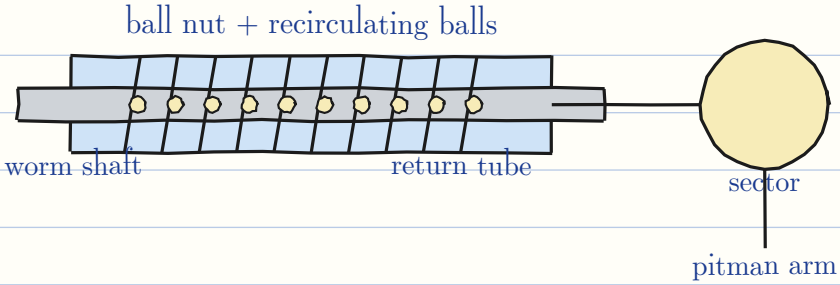


Fig 3.8 Recirculating-ball steering box

Rack & pinion steering gear

- Pinion on the steering shaft meshes with a toothed rack; rotation of the pinion slides the rack left / right; tie rods at the ends move the wheel arms directly -> no pitman arm, drag link, idler.
- Advantages: simple, compact, light, low cost, positive and quick response, good road feel, fewer joints. Disadvantages: transmits road shocks / kick-back, not suitable for heavy axle loads; can be damaged by impact.
- Used in almost all modern small & medium cars (MacPherson suspension); power-assisted by hydraulic or electric motor.

Point	Recirculating ball	Rack & pinion
Parts	More (box, arm, links)	Few
Feel	Isolated, less feedback	Direct, sporty
Load capacity	High	Moderate
Use	Trucks, jeeps	Cars

9. Power Steering

- Power steering supplements the driver effort with hydraulic or electric force -> light steering at parking speeds, still gives feel at speed. If assist fails, the vehicle can still be steered manually (fail-safe).

Types

- Hydraulic power steering (HPS) - engine-driven vane pump (power-assist 70-100 bar), fluid reservoir, rotary control valve, power piston / cylinder, hoses.
- Electro-hydraulic (EHPS) - pump driven by electric motor (saves fuel).
- Electric power steering (EPS) - electric motor on column / rack; torque sensor + ECU (see next page).

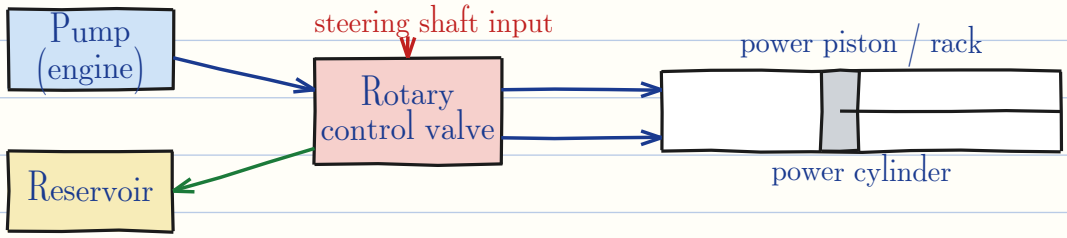


Fig 3.9 Hydraulic power steering (integral rack type)

Working

- When the driver turns the wheel, a torsion bar in the rotary valve twists a little -> valve spool opens oil passage to one side of the power piston -> hydraulic force assists the rack; the other side is open to return. When the wheel stops, the torsion bar un-twists, valve centres, assist stops.
- Types by layout: integral (gear + piston in one housing), linkage (cylinder separate), semi-integral.
- Fluid: ATF / special PSF; faults - belt slip, low fluid (groan), air in system, leaking seals / hoses.

10. Electric Power Steering (EPS)

- EPS uses an electric motor (brushed / brushless) to assist steering, controlled by an ECU using signals of steering torque, steering angle and vehicle speed.

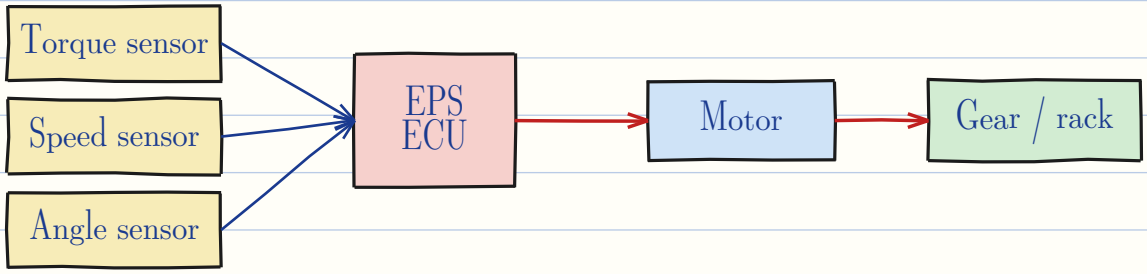


Fig 3.10 Electric power steering (block diagram)

Types by motor location

- Column-assist (C-EPS) small cars; pinion-assist (P-EPS); rack-assist (R-EPS) - bigger vehicles & high loads.

Advantages

- No hydraulic fluid, pump, hoses -> compact, eco-friendly; assist only when needed -> saves 3-5 % fuel; variable assist (heavy at high speed); supports lane-keep, park assist, self-parking; tunable feel.

Disadvantages

- Less natural road feel (improved with software), needs high current (vehicle electrical load), cost for heavy-duty vehicles.

Point	Hydraulic PS	Electric PS
Power source	Engine-driven pump	Electric motor (battery)
Efficiency	Pump runs always (parasitic loss)	On demand only
Maintenance	Fluid, belts, hoses	Almost none
Tuning	Fixed	Software adjustable

Steer-by-wire

Future: no mechanical link - sensors + actuators (steer-by-wire) with redundant ECUs; used in advanced / autonomous prototypes.

11. Front Axle & Stub Axles

- Front axle carries the vehicle weight at the front, resists braking & steering forces, and holds the wheels via stub axles that pivot on king pins.

Types

- Live front axle - also drives wheels (4WD / AWD vehicles). Dead front axle - only supports weight and steering (commonly used).
- Section: I-section / H-section forged steel (medium-carbon, Ni-Cr steel) - drop-forged I-beam is the usual choice; tubular for light vehicles. Ends are shaped for the king-pin boss; centre is dropped to lower the engine.

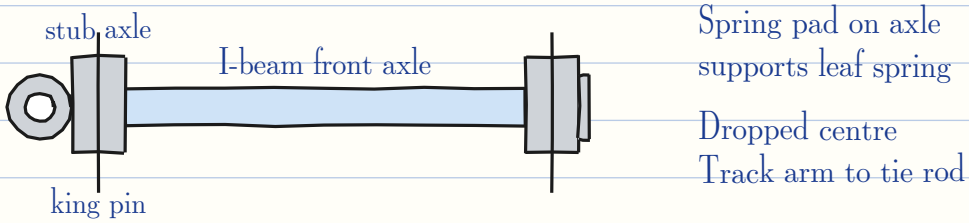


Fig 3.11 Front axle with stub axles

Stub axle arrangements (how the king-pin is connected)

Type	Description
Elliot	Yoke (fork) on the axle end holds stub axle; king-pin fixed in the axle; stub axle has the bearing
Reverse Elliot	Yoke is on the stub axle, axle end fits in between; most common
Lamoine	Stub axle L-shaped, king-pin in axle end, fork on stub axle (opening downwards)
Reverse Lamoine	L-shape stub axle, inverted; king-pin in stub axle

- King-pin bushed with bronze bush; thrust bearing (roller / ball) takes the vertical load; shims set vertical play.

12. Suspension System - Introduction

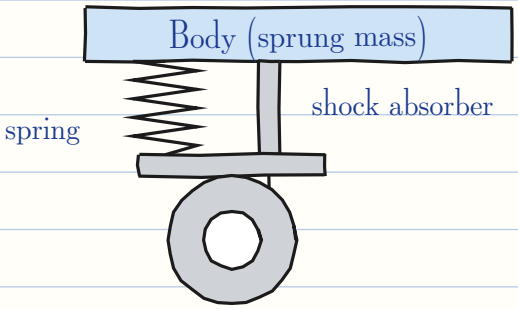
- Suspension = system of springs, shock absorbers and linkages connecting the vehicle body (sprung mass) to the wheels (unsprung mass).

Functions

- Isolate body from road shocks (ride comfort); keep tyres in contact with road (road holding); support weight; maintain correct wheel alignment; resist roll, pitch, dive & squat; transmit driving & braking forces.

Parts

- Springs (leaf, coil, torsion bar, air, rubber), dampers (shock absorbers), control arms / wishbones, links / rods, anti-roll bar, bushes and ball joints.



Sprung mass: body, engine, frame

Unsprung mass: wheels, tyres, axle, brakes -> should be minimum

Sprung / unsprung ratio high = better ride

Fig 3.12 Basic quarter-car suspension

Classification

- Rigid (solid / dependent) axle - both wheels on one axle: leaf spring, coil spring, air; Independent - each wheel moves separately: wishbone, MacPherson, trailing arm, swing axle, multi-link.
- Based on spring: leaf, coil, torsion bar, air, hydro-pneumatic, rubber. Active/semi-active electronic suspension (Unit V).
- Natural frequency for comfort: 1 - 1.5 Hz (60 - 90 cycles/min).

13. Independent Suspension

- Each wheel is connected to the body separately, so a bump on one side does not affect the other wheel.

Advantages

- Better ride & handling; lower unsprung weight; more engine space (no axle); less wheel shimmy; easy to design geometry.

Disadvantages

- Complex, costly, more joints; track & camber change with wheel travel.

Common types

1. Double wishbone (SLA) - upper + lower arms of unequal length; coil spring + damper; good camber control; sports cars, SUVs.
2. MacPherson strut - strut (spring + damper + steering pivot) with a single lower control arm; simple, light, saves space; most FWD cars.
3. Trailing arm / semi-trailing arm - rear suspension; arm pivots at front, carries wheel at rear.
4. Swing axle - half-axles pivoted at differential (VW Beetle); big camber change.
5. Multi-link - 3-5 links per wheel; best compromise comfort/handling; luxury cars.

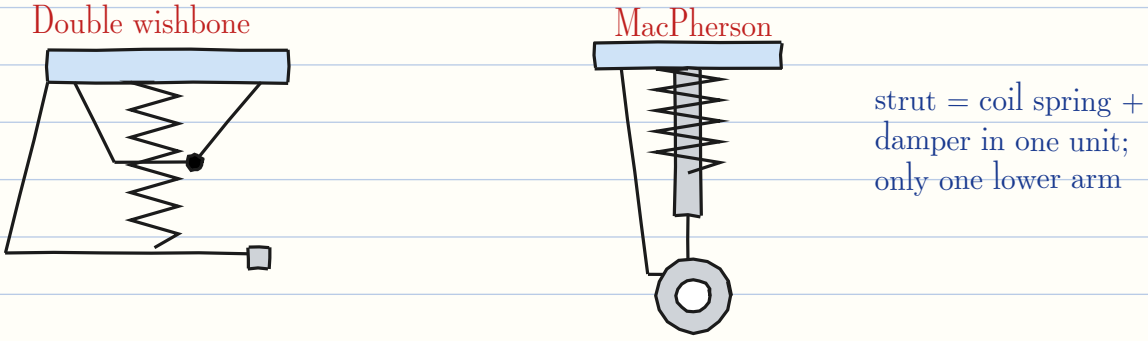


Fig 3.13 Wishbone and MacPherson strut (front view)

14. Solid Axle & Leaf Spring Suspension

- Solid (rigid) axle suspension: both wheels on one beam axle (front or rear), attached to the frame via springs. Used in trucks, buses, jeeps, SUV rear axles.
- Advantages: simple, strong, low cost, camber constant with load, high load capacity. Disadvantages: higher unsprung mass, bump on one wheel tilts the other; poorer ride.

Leaf spring

- A stack of steel plates (leaves) of decreasing length clamped together: semi-elliptic is the commonest form. Longest = master (main) leaf with eyes at both ends (front eye fixed on pin / rear on shackle to allow length change).

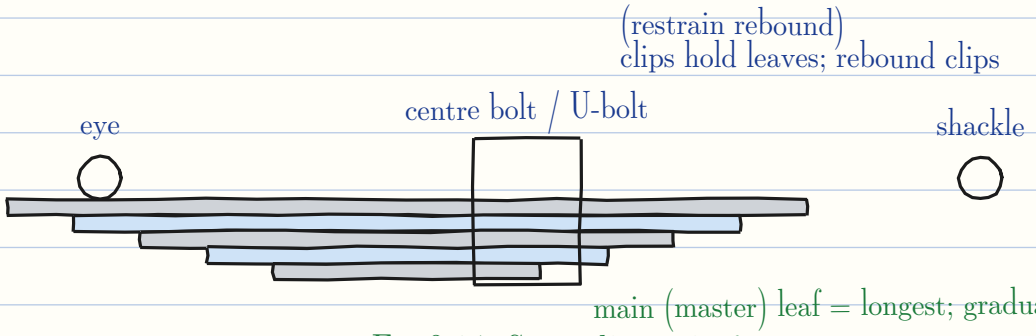


Fig 3.14 Semi-elliptic leaf spring

- Main parts: master leaf, graduated leaves, centre bolt, rebound clips, U-bolts to axle, spring seat, shackle & bushes, inter-leaf friction (damps vibrations, but causes harshness -> use lubricated / plastic liners).
- Material: silico-manganese / chrome-vanadium steel, shot-peened for fatigue life. Types: semi-elliptic, quarter-elliptic, transverse, parabolic (taper) leaf - fewer leaves, lighter, less friction.
- Spring rate k = load / deflection; leaf spring also locates the axle and transmits drive / braking torque (no extra links needed).

15. Coil Spring & Torsion Bar

Coil spring

- Helically wound round steel wire (spring steel, 10-20 mm dia.). Stores energy by torsion of wire. Used in independent front & rear suspension (car, SUV).
- Advantages: lightweight, compact, no inter-leaf friction, durable, wide rate range, progressive-rate (variable pitch / conical) possible. Disadvantage: cannot take side loads -> needs control arms / links.
- Rate: k = G d^4 / (8 D^3 n) (G = shear modulus, d = wire dia, D = mean coil dia, n = active coils).

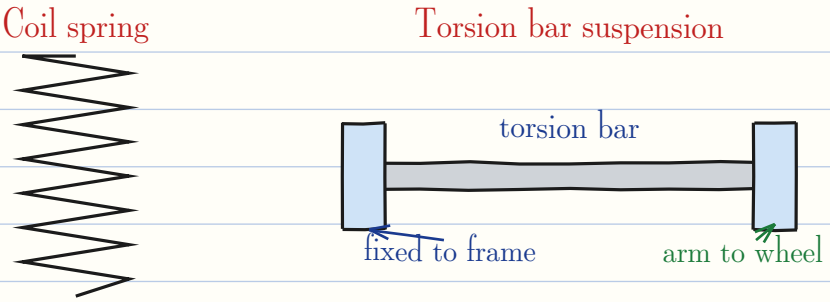


Fig 3.15 Coil spring and torsion bar

Torsion bar spring

- A steel rod fixed at one end to the frame, other end connected to a lower control arm; wheel movement twists the bar and its elasticity gives spring action.
- Pros: compact, adjustable ride height by turning the anchor, light, long life, low cost. Cons: limited deflection (twist), needs well-finished surface; left/right bars not interchangeable (pre-stressed).
- Used in Citroen, VW Beetle, Chrysler (front), many 4WD pickups & Military vehicles.

Rubber spring

- Rubber blocks in compression / shear - give self damping, noise insulation. Used as auxiliary / bump stops, engine mounts, bushes, trailers.

Spring rate & frequency

f = (1/2pi) sqrt(k / m) . Softer spring (low k) gives lower frequency = more comfort but more body roll; stiffer spring = better handling.

16. Air Suspension

- Uses air bags (bellows / air springs) instead of steel springs. Compressed air (4 - 8 bar) carries the load; pressure is varied to control ride height & stiffness.

Main components

- Compressor, dryer + reservoir, height (levelling) valve (linkage to axle), air spring (rolling lobe / bellows / convoluted type), shock absorbers, control ECU with ride-height sensors, solenoid valves.



Load increases -> body sinks -> valve admits air -> body rises (self levelling)

Fig 3.16 Air suspension layout

Advantages

- Constant ride height & frequency irrespective of load; excellent comfort; adjustable (highway / off-road / loading mode); long life of cargo; reduces road damage; used in buses, trucks, trailers, Volvo, Mercedes, Range Rover, Audi.

Disadvantages

- Costly, complex, air leaks, needs compressor & maintenance; limited lateral stiffness (needs linkages).

Types of air springs

- Bellows (convoluted) - 1 to 3 folds, high load capacity, used on trucks / buses. Reversible sleeve - flexible sleeve rolls over a piston. Rolling lobe - compact, long stroke, used in cars (smooth ride).
- Air-over-oil / hydropneumatic (Citroen) - gas spring with oil damper & pump; self-levelling with very soft ride.

Working idea

Spring rate rises automatically with load because the air pressure rises - so the natural frequency stays almost constant from empty to full load.

17. Shock Absorbers & Stabiliser Bar

- Shock absorber (damper) dissipates the spring's stored energy as heat to damp oscillations; without it the body bounces repeatedly. Controls spring rebound & wheel hop.
- Types: friction (old), hydraulic telescopic (double acting) - standard, gas-filled (monotube / twin-tube), adjustable, magnetorheological (Unit V).

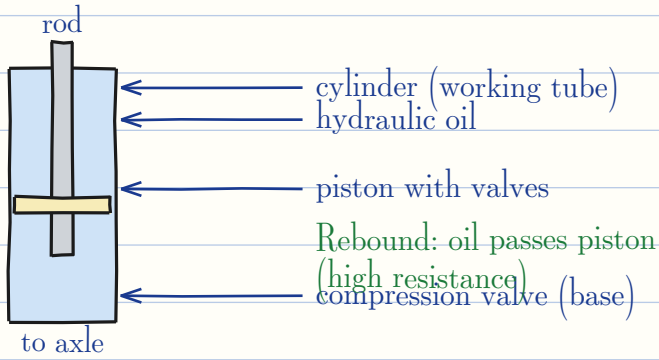


Fig 3.17 Telescopic hydraulic damper (twin tube)

- Working: on compression (bump) oil is forced through small orifices / the base valve; on rebound the piston valve restricts oil flow more (rebound damping ~ 2-3 x compression damping). Energy -> heat -> dissipated through the tube.
- Gas-filled (nitrogen 5-25 bar): prevents oil foaming (aeration) and fade; better response.

Stabiliser (anti-roll) bar

- U-shaped torsion bar across axle linking left & right lower arms; resists body roll in corners by twisting (equalises load on both wheels) without affecting ride in straight line.

Spring	Typical use
Leaf	Trucks, buses, rear of vans / jeeps
Coil	Cars - independent
Torsion bar	Compact 4WD, some cars
Air	Buses, luxury cars, trailers